

# Yi-Chun Liao

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## EDUCATION

### Duke University

*Ph.D. in Electrical and Computer Engineering*

- Advisor: *Prof. Yiran Chen*

**Durham, NC**

*Aug. 2026 – Present*

### National Taiwan University (NTU)

*B.Sc. in Computer Science and Information Engineering (CSIE)*

- Overall GPA: 3.90/4.30, Last 60 GPA: 4.09/4.30

**Taipei, Taiwan**

*Sep. 2021 – Jan. 2026*

## PUBLICATIONS

**Yi-Chun Liao**, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “RETENTION: Resource-Efficient Tree-Based Ensemble Model Acceleration with Content-Addressable Memory,” in *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, doi: [10.1109/TCAD.2026.3691288](https://doi.org/10.1109/TCAD.2026.3691288).

**Yi-Chun Liao**, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “Practicalizing Tree-Based Model Acceleration with CAM through Model Pruning and Data Placement Optimization.” *ACM/IEEE International Conference on Hardware/Software Codesign and System Synthesis (CODES+ISSS '25)*, September 28–October 3, 2025, Taipei, Taiwan. ACM, New York, NY, USA, 2 pages. doi:[10.1145/3742873.3758342](https://doi.org/10.1145/3742873.3758342).

Hong-Wei Li, Ping-Ting Liu, Bo-Wei Lin, **Yi-Chun Liao**, and Yennun Huang. “IPMES: A Tool for Incremental TTP Detection Over the System Audit Event Stream,” *2024 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, Brisbane, Australia, 2024, pp. 265–273, doi:[10.1109/DSN58291.2024.00036](https://doi.org/10.1109/DSN58291.2024.00036).

## RESEARCH EXPERIENCE

### Research Intern

*University of Notre Dame, Advisor: Prof. Xiaobo Sharon Hu*

**Indiana, United States**

*May 2025 – Present*

- Conduct independent research on realizing scalable in-memory retrieval for retrieval-augmented generation (RAG), co-designing algorithm and hardware to prevent the massive data movement of retrieval phase.
- Conduct independent research on alleviating the capacity constraint of in-memory hyperdimensional computing, adapting the model for hardware constraints and introducing an algorithm for scalable CAM implementations.
- Contribute to the extension of the open-source CAM simulator [CAMASim](#), enabling data mapping optimizations with SPC and ODR, algorithms proposed in RETENTION.

### Research Assistant

*Academia Sinica, Advisors: Prof. Yuan-Hao Chang and Prof. Tei-Wei Kuo*

**Taipei, Taiwan**

*Apr. 2024 – Jun. 2026*

- Conducted independent research on minimizing resource consumption for tree-based ensemble model inference, introducing an end-to-end framework [RETENTION](#) to minimize the capacity requirement for CAM-based acceleration.
- Explored and quantified the relationship between capacity redundancy and processing overhead, enabling flexible tradeoffs between controllable accuracy loss, energy consumption, and capacity requirement.
- Reduced capacity requirement by  $4.35\times$  to  $207.12\times$  with less than 3% accuracy loss and no degradation on throughput and energy efficiency. The first-authored paper was published in TCAD.
- Collaborated on minimizing retrieved tokens for RAG acceleration, exploring the tradeoffs between accuracy, latency, and capacity requirement.

### Research Assistant

*Academia Sinica, Advisor: Prof. Yennun Huang*

**Taipei, Taiwan**

*Jul. 2023 – Jun. 2024*

- Collaborated on developing intrusion detection tool [IPMES](#) to detect incremental tactics, techniques, and procedures (TTPs) in system audit event streams. The co-authored paper was published in DSN 2024.
- Constructed an automated system independently to record hundreds of system attacks, utilizing [MITRE Caldera](#) to inject malware and TTPs into virtual machines. The system was subsequently used for synthesizing system attack datasets.

# TEACHING EXPERIENCE

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**Teaching Assistant**

Taipei, Taiwan

Computer Architecture, Lecturer: *Prof. Chia-Lin Yang*

Sep. 2024 – Jan. 2025

- Supported the instruction of the junior compulsory course Computer Architecture at NTU, enrolled by over 100 students.
- Led weekly discussion sessions, designed and/or graded assignments, quizzes, and exams.
- Provided voluntary office hours, exceeding 10 hours per week during exam periods.

**Teaching Assistant**

Taipei, Taiwan

APCS Camp

Jul. 2022/23 – Aug. 2022/23

- Provided TA hours for over 300 C++ beginners during two-week crash courses.
- Adapted teaching strategies including programming problem solving and customized course reviews to meet the diverse needs and skill levels of students.

# HONORS

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**First Place of 2025 Bachelor Thesis Award**

Taipei, Taiwan

National Taiwan University

Jun. 2025

- Distinguished through a three-stage selection process and recognized as one of the top 2 in the CSIE Department, the best in the EECS College, and one of the top 6 university-wide.

**Appier Enterprise Award**

Taipei, Taiwan

National Taiwan University

Jun. 2025

- Received the Most Potential Award from Appier in the departmental undergraduate research exhibition.

**Dean’s List Award**

Taipei, Taiwan

National Taiwan University

Mar. 2025

- Ranked 1/155 in the department for both Spring and Fall 2024 semesters.

**Irving T. Ho Memorial Scholarship**

Taipei, Taiwan

National Taiwan University

Feb. 2025

- Awarded to 3 students in the department for academic and research excellence.

**First Place of AUO Enterprise Award**

Taipei, Taiwan

MakeNTU 2024 (Nationalwide Makerthon)

May 2024

- Designed an interactive in-car RPG game for AUO smart windows to enhance travel experience, which was recognized as the best work for AUO Enterprise Award among 43 teams.

# EXTRACURRICULAR ACTIVITIES

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**President**

Taipei, Taiwan

NTU CSIE Department Student Association

Aug. 2023 – Jul. 2024

- Directed the student council of the CSIE department with over 50 members.
- Promoted student welfare and represented over 600 students in faculty meetings, contributing to departmental decisions.

**Vice President**

Taipei, Taiwan

NTU EECS College Student Association

Aug. 2023 – Jul. 2024

- Represented over 1500 students in college faculty meetings and helped shape student-related policies.

**President**

Taipei, Taiwan

NTU Cancer Children Service Club

Feb. 2023 – Jul. 2023

- Led a 50-member service club, managing budget allocation and arranging educational and recreational activities.
- Worked with Childhood Cancer Foundation of R.O.C., organizing voluntary services to reduce the burden of parents.

**Vice Coordinator**

Taipei, Taiwan

NTU CSIE Camp

Feb. 2023 – Jul. 2023

- Organized a 6-day camp for over 100 high school students with a 50-member staff team.
- Designed activities to introduce students to the CSIE department and provide insights into university life.